Supply Chain Visibility & Optimisation

Milestone 2: Inventory Data Analysis & Power BI Dashboard Development

Product Movement, Sales, Delivery Performance, Regional Drilldown and Variance Analysis

Project Detail	Information
Project Type	Team Project
Team Number	4
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Internship / Program	Infosys Springboard Virtual Internship 7.0 - Batch 2
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Documentation Scope

This document explains the final Milestone 2 Power BI report and uses the current PBIP project, cleaned CSV, semantic model, and latest five-page PDF export as the source of truth.

Table of Contents

This structured contents page lists the major sections included in the Milestone 2 documentation.

1. Milestone 2 Overview

Milestone 2 transforms the cleaned Olist Brazilian E-Commerce dataset into an interactive Power BI Supply Chain Control Tower. The report covers executive performance, product movement used as an inventory proxy, sales analysis, delivery reliability, regional/product drilldown, trends, and month-over-month variance.

The mentor checklist required inventory performance, delivery performance, region and product drilldown, trends, and variance analysis. The final report addresses these through five Power BI pages: Executive Supply Chain Overview, Inventory Movement Intelligence, Delivery Performance, Region & Product Drilldown, and Trends & Variance Analysis.

2. Dataset Context

The project uses a cleaned Olist Brazilian E-Commerce dataset containing 66,039 rows. The dataset combines order, item, product, customer, seller, payment, review, freight, and delivery information.

Analytical Area	Representative Fields Used
Order and item grain	order_id, order_item_id, Movement Line Key
Product analysis	product_id, product_category_name, product_category_name_english, Product Category Clean
Sales and freight	price, freight_value
Date analysis	order_purchase_timestamp, Order Date, Date[Month Start], Date[Month Year]
Delivery analysis	order_status, order_delivered_customer_date, order_estimated_delivery_date, delivery_delay
Location analysis	customer_state, customer_city, seller_state, seller_city

3. Important Inventory-Proxy Methodology

Mandatory methodology statement
Product movements are used as an inventory proxy; the dataset does not contain actual warehouse stock or on-hand inventory.

The Olist dataset does not provide warehouse stock balances, stock snapshots, reorder points, or on-hand inventory. Therefore, Milestone 2 avoids claiming actual inventory availability. Inventory-oriented analysis is based on observed order-item movement.

- Units Moved
- Active Products
- Product Velocity
- Product Movement Value
- ABC Classification
- Fast/Slow movement classification
- Demand Variability
- Category Movement
- Top Products by Movement

4. Data Grain and Duplicate-Safe Sales Logic

The merged dataset can repeat order-item rows when an order has multiple payment records. A direct SUM(price) could therefore overstate sales. The report uses the unique order-item grain defined by order_id + order_item_id, implemented as Movement Line Key.

Validation Item	Value
CSV rows	66,039
Unique order_id + order_item_id	63,089
Raw SUM(price)	4,280,314.80
Safe item-grain Total Sales	4,097,737.13
Raw inflation factor	approximately 1.0446x

Implemented DAX measure:

```
Total Sales =
SUMX(
    VALUES('Supply_Chain_Cleaned_Dataset'[Movement Line Key]),
    CALCULATE(MAX('Supply_Chain_Cleaned_Dataset'[price]))
)
```

In business terms, the measure takes each unique order item once and sums its item price once. Freight is not included in Total Sales.

5. Power BI Data Model

The semantic model uses the main fact table, a calculated Date table, and a Product Intelligence calculated table. The Date table supports chronological month analysis, and Product Intelligence supports ABC and movement classification by product.

Model Component	Role
Supply_Chain_Cleaned_Dataset	Main table containing orders, order items, product, freight, delivery, customer, seller, payment, and review attributes.
Date	Calculated calendar table built from Order Date with Year, Quarter, Month, Month Start, Month Year, Day, and Date hierarchy.
Product Intelligence	Calculated product-level table with product_id, product category, Units Moved, Movement Value, Movement Rank, Cumulative Value %, ABC Classification, and FSN Classification.
Relationships	Order Date relates to Date[Date]; product_id relates to Product Intelligence[product_id].
Customer hierarchy	Customer Region -> State -> City -> Product Category -> Product.
Seller hierarchy	Seller State -> Seller City -> Seller -> Product Category -> Product. Sellers are not treated as warehouses.

6. Key KPI Definitions

KPI	Purpose	Business Meaning
Total Orders	Measures distinct customer orders.	Overall order volume across the dataset.
Total Sales	Measures item sales safely at unique order-item grain.	Actual item sales value excluding freight, protected from duplicate payment-row inflation.
Units Moved	Counts distinct Movement Line Key values.	Observed order-item movement used as the inventory proxy.
Product Movement Value	Sums item price once per Movement Line Key.	Value of product movement used for inventory-proxy analysis.
Active Products	Counts distinct product_id values.	Breadth of products observed in order history.
Product Velocity	Units Moved / (Active Products x Active Movement Months).	Average movement rate per active product per active month.
Demand Variability	Monthly movement standard deviation divided by monthly average.	Relative volatility in monthly product movement.
On-Time Delivery %	On-time delivered orders divided by delivered orders.	Orders delivered on or before the estimated delivery date.
Late Delivery %	Late delivered orders divided by delivered orders.	Share of delivered orders arriving after the estimated delivery date.
Average Delivery Time	Average days from purchase to customer delivery.	Typical end-to-end delivery duration.
Average Delivery Variance	Average delivery_delay in days.	Negative means early delivery; positive means late delivery.
Average Early Delivery Days	Average early days for orders delivered before estimate.	Typical early-delivery margin.
Average Late Delivery Days	Average late days for delayed orders.	Typical delay size for late deliveries.
Freight Cost	Sums freight_value once per Movement Line Key.	Logistics cost associated with observed product movement.
MoM Variance	Compares latest/current month with the immediately previous month.	Highlights directional change in movement, value, freight, and on-time performance.

7. Product Movement Intelligence

ABC Classification is based on cumulative Product Movement Value contribution. Class A covers products contributing up to 80% of cumulative value, Class B covers more than 80% and up to 95%, and Class C covers the remaining products above 95%. This separates high-value product movement from lower-value long-tail movement.

Movement classification is calculated from observed order-history movement frequency. The model defines Fast products at or above the 67th percentile of units moved, Slow products at or below the 33rd percentile, and Moderate products between those thresholds. The final visible report output currently displays Fast and Slow categories; no Non-Moving class is assigned because the dataset only contains observed order history.

Product Velocity is documented in the model as average units moved per active product per active month: Units Moved / (Active Products x Active Movement Months). Demand Variability is calculated as the standard deviation of monthly movement divided by the average monthly movement.

8. Dashboard 1 - Executive Supply Chain Overview

Figure 1. Executive Supply Chain Overview dashboard page.

This flagship page gives management a concise view of order volume, movement, sales, delivery reliability, and regional/product performance.

Displayed KPI / Visual	Explanation
Total Orders: approximately 56K	Distinct customer order count.
Units Moved: approximately 63K	Distinct order-item movement lines.
Product Movement Value: approximately 4.10M	Movement value used for inventory-proxy analysis.
On-Time Delivery: 93.91%	Delivered on or before the estimated date.
Total Sales: approximately 4.10M	Duplicate-safe sales at unique order-item grain.
Average Delivery Time: 11.52 days	Average purchase-to-delivery duration.
Product Movement Value Trend	Shows growth in product movement value over time.
Delivery Status Mix	Compares on-time/early and late delivered orders.
Orders by Product Category	Top categories by distinct order volume.
Sales by Product Category	Top categories by duplicate-safe Total Sales.
Units Moved by Region	Shows Southeast as the dominant movement region.

9. Dashboard 2 - Inventory Movement Intelligence

Figure 2. Inventory Movement Intelligence dashboard page.

This page is the product-movement command centre. It supports inventory-proxy analysis without claiming actual warehouse stock. It displays Units Moved, Active Products, Product Velocity, Product Movement Value, and Demand Variability.

The page includes Category Movement, ABC Classification Mix, Movement Classification Mix, Top 10 Products by Movement, and Product Intelligence Detail. The Top 10 Products visual uses Product Category plus a shortened/anonymized product_id because the Olist dataset does not provide commercial product names for individual products.

10. Dashboard 3 - Delivery Performance

Figure 3. Delivery Performance dashboard page.

This page focuses on operational reliability. It tracks whether delivered orders arrived on time or late, how delivery performance changes over time, and which regions or cities show delivery pressure.

Metric / Visual	Interpretation
On-Time Delivery % = 93.91%	Most delivered orders arrived on or before the estimated date.
Late Delivery % = 6.09%	Late orders are a relatively small share of delivered orders.
Average Delivery Time = 11.52 days	Average duration from purchase to customer delivery.
Average Delivery Variance = -11.81 days	Negative values indicate deliveries were early on average, not delayed.
Average Early Delivery = 13.42 days	Average number of days early for early deliveries.
Average Late Delivery = 10.14 days	Average delay size for late deliveries.
On-Time Delivery Trend	Shows monthly reliability changes.
Delivery Severity	Groups deliveries into Early, On Time, 1-3 Days Late, 4-7 Days Late, 8+ Days Late, and Unknown.
Late Orders by Region	Compares late-order concentration by customer region.
Average Delivery Time by City	Uses the implemented 10+ delivered orders threshold to avoid misleading one-order city comparisons.

11. Dashboard 4 - Region & Product Drilldown

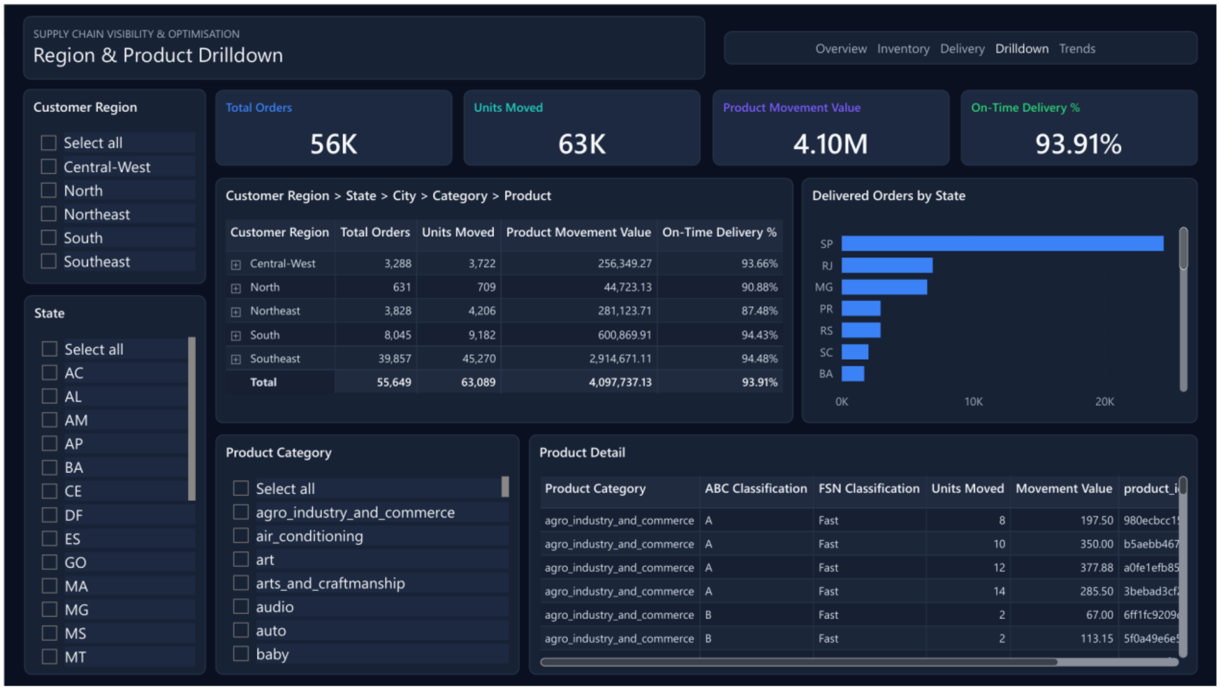

Figure 4. Region & Product Drilldown dashboard page.

This page provides an investigative drilldown workspace. The main hierarchy is Customer Region -> State -> City -> Product Category -> Product. Users can move from high-level regional performance to increasingly detailed geographic and product-level performance.

The page keeps summary KPIs visible while drilling: Total Orders, Units Moved, Product Movement Value, and On-Time Delivery %. It also includes Delivered Orders by State, Product Detail, Customer Region, State, and Product Category slicers.

12. Dashboard 5 - Trends & Variance Analysis

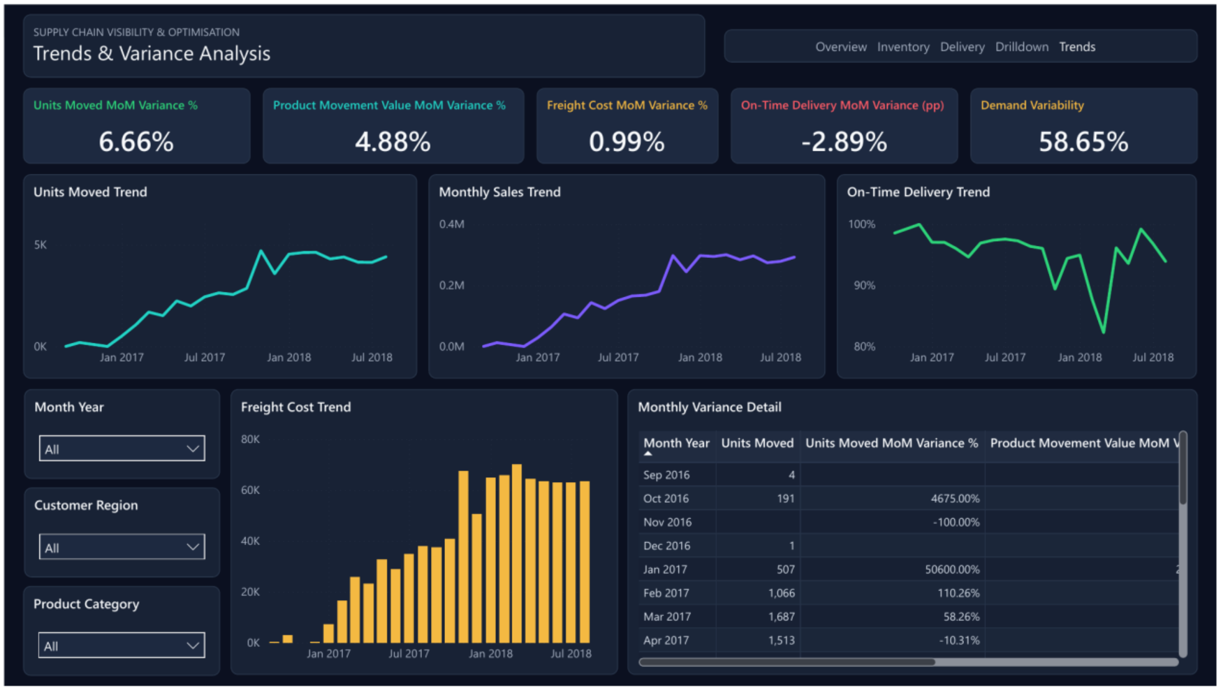

Figure 5. Trends & Variance Analysis dashboard page.

This page monitors monthly performance changes. KPI cards show the latest visible month's variance when no specific month is selected.

Displayed KPI / Visual	**Explanation**
Units Moved MoM Variance % = approximately +6.66%	Latest visible month movement change versus the previous month.
Product Movement Value MoM Variance % = approximately +4.88%	Latest visible movement-value change versus the previous month.
Freight Cost MoM Variance % = approximately +0.99%	Latest visible freight-cost change; cost increases are treated as cautionary.
On-Time Delivery MoM Variance = approximately -2.89 percentage points	Percentage-point change, not percentage-growth logic.
Demand Variability = approximately 58.65%	Relative month-to-month movement volatility.
Units Moved Trend	Chronological units moved over time.
Monthly Sales Trend	Duplicate-safe Total Sales over time.
On-Time Delivery Trend	Delivery reliability over time.
Freight Cost Trend	Freight cost over time.
Monthly Variance Detail	Shows monthly current and variance values.

Very large early-month percentage changes, such as 4,675% or 50,600%, are low-base effects caused by very small prior-month volumes. They are not calculation errors.

13. Month-over-Month Variance Analysis

Month-over-month variance compares a current month with the immediately previous calendar month. The Date table provides Month Start and Month Year fields so trends remain chronological. Previous-month measures identify either the selected month or the latest visible month with data, calculate the previous month using EDATE, and return blank when no previous-month data exists.

1. Current Month: the selected Month Start / Month Year, or the latest visible month with data.
2. Previous Month: one calendar month before the target month.
3. Variance: Current Month minus Previous Month.
4. Variance %: absolute variance divided by Previous Month, except On-Time Delivery MoM, which is a percentage-point difference.

14. Key Business Insights

- Overall on-time/early delivery performance is approximately 93.91%, while late deliveries represent approximately 6.09%.
- Southeast is the dominant region in product movement and delivered-order state volume.
- Total Sales are approximately 4.10M at the validated unique order-item grain.
- Approximately 63K units moved across approximately 18K active products.
- Demand Variability is approximately 58.65%, indicating material movement volatility across months.
- Product movement and sales increase materially over the observed period.
- Freight costs rise with supply-chain activity, making freight trend monitoring important.
- ABC classification separates high-value movement contributors from lower-value groups.
- Movement classification identifies fast and slow products in the final visible report output.

15. Mentor Requirement Coverage

Requirement	Implementation	Status
Total Orders	Executive KPI card	Covered
Total Sales	Executive KPI card and Monthly Sales Trend	Covered
On-Time %	Executive and Delivery KPI cards	Covered
Average Delivery Time	Executive and Delivery KPI cards	Covered
Product Movement / Inventory Proxy	Inventory Movement Intelligence methodology and KPIs	Covered
Product Movement by Category	Category Movement visual	Covered
Top 10 Products by Movement	Inventory Movement horizontal bar chart	Covered
Product Movement Trend	Executive trend visual	Covered
ABC Classification	ABC Classification Mix and Product Intelligence table	Covered
Movement Classification	Movement Classification Mix and Product Intelligence table	Covered
Demand Variability	Inventory and Trends KPI cards	Covered
Sales by Product Category	Executive horizontal bar chart	Covered
Product Performance	Product Intelligence Detail and Product Detail tables	Covered
Monthly Sales Trend	Trends & Variance line chart	Covered

Orders by Product Category	Executive horizontal bar chart	Covered
On-Time vs Late	Delivery Status Mix and Delivery Performance KPIs	Covered
Delivered Orders by State	Region & Product Drilldown bar chart	Covered
Average Delivery Time by City	Delivery Performance bar chart	Covered
State -> City -> Product Drilldown	Customer Region -> State -> City -> Product Category -> Product hierarchy	Covered
Monthly trends, MoM variance, freight trend, on-time trend	Trends & Variance Analysis page	Covered

16. Challenges and Solutions

Challenge	Solution
No real warehouse stock data	Used product movements as an inventory proxy and documented that no stock-on-hand is available.
Duplicate-looking rows caused by multiple payment records	Used Movement Line Key / unique order-item grain for safe sales and movement measures.
No commercial product names	Used Product Category plus shortened product_id for product-level presentation.
Previous Month / MoM DAX issue	Previous-month measures were validated and corrected using Date-table filtering and latest-visible-month logic.
Long technical labels and readability	Used presentation-friendly visual titles and shortened product display labels.

17. Limitations

- The dataset does not contain warehouse stock-on-hand or inventory snapshots.
- True average inventory balance and true inventory turnover cannot be calculated.
- Product movement is a proxy based on observed order history, not actual warehouse availability.
- Commercial product names are anonymized or unavailable; product_id is used for detailed product analysis.
- The analysis is historical and does not forecast future demand or delivery performance.
- Very small monthly base values can create extreme percentage variance.

18. Business Value

The final dashboard improves supply-chain visibility by bringing order, sales, product movement, delivery, freight, geography, and variance measures into one coherent control-tower report.

- Management can monitor executive KPIs and detect high-level performance shifts.
- Product teams can identify high-value, fast-moving, and slow-moving products.
- Operations teams can evaluate delivery reliability and severity patterns.
- Regional teams can drill from region to state, city, category, and product.
- Finance and logistics teams can monitor sales, freight costs, and movement-value trends.
- Variance analysis highlights monthly changes requiring attention.

19. Conclusion

Milestone 2 converted the cleaned Olist dataset into a professional interactive Power BI Supply Chain Control Tower. The report covers executive KPIs, duplicate-safe sales, product movement used as an inventory proxy, delivery performance, geographical drilldown, monthly trends, and variance analysis.

The dashboard satisfies the mentor checklist while remaining transparent about data limitations. It provides a strong analytical foundation for subsequent project milestones, portfolio presentation, and mentor review.

20. Final Validation Summary

Validation Item	Result
Final report pages	Five pages verified from latest PDF export.
KPI values	Verified against dashboard and dataset/model logic.
Total Sales logic	Verified as duplicate-safe at Movement Line Key grain.
Inventory-proxy wording	Included exactly and consistently.
On-Time / Late terminology	On-time includes orders delivered on or before estimated date.
ABC rules	Documented as cumulative movement-value thresholds: A <= 80%, B <= 95%, C > 95%.
Movement classification	Documented from implemented percentile rules and final visible classes.
Hierarchy	Customer Region -> State -> City -> Product Category -> Product verified.
Mentor checklist	All listed items marked Covered.

End of the Documentation for Milestone 2